

[Max Marks: 30]

[Duration: 90 minutes]

IIT Guwahati - Department of Computer Science & Engineering

CS 223- Computer Organization & Architecture – Quiz 2 (06.03.2021)

Submission Link:

<https://forms.office.com/Pages/ResponsePage.aspx?id=jacKheGUxkuc84wRtTBwHANSILeDKUNOi1YB2VJD6t1UOTdUMVVYUlk4WFRUOEw0Q0dZVEdHV0tSNy4u>

Answer all the questions. Mark the correct question number. Show calculations and diagrams wherever necessary. 6 marks each for first 3 questions and 4 marks each for next 3 questions.

1. Consider the following MIPS code snippet that is part of a large program. What are the kind of hazards possible here? How can you reduce/eliminate them?

MUL.D	F1, F2, F3
ADD.D	F4, F5, F6
ADD.D	F7, F4, F8
L.D	F4, 8(R0)
ADD.D	F8, F4, F1

2. In a program there is a branch instruction which is iterated 6 times. The processor uses a (2, 2) correlating branch predictor. Each entry of BHT is a 2-bit value that is updated by a standard 2-bit, 4-state finite state machine. Let the initial entry of the BHT for NN/NT/TN/TT is 00/00/11/11, respectively. At the end of prediction for 6 instances of branches and subsequent state update after branch outcome, the BHT for NN/NT/TN/TT is 01/01/10/10, respectively. The branch predictor has an accuracy of 0.333. The BHT is indexed with an NN value initially. Illustrate the operation of the branch predictor with the help of a neat labelled BHT.
 - (a) List the actual outcomes of the 6 iterations of this branch instruction in order.
 - (b) How many times the predictor changes its state?

3. Design a 1024 X 8 memory using 256 X 2 chips.
 - (a) How many chips will be required row wise and column wise?
 - (b) Show the design layout with all the necessary units including decoder, MAR and MBR.

4. ABC company advertised that their magnetic disk has the following specification: Rotational speed: 3600 rpm, Average seek time: 10ms, Sector size :2KB, track to track delay 1.5ms. Number of surface: 8, tracks per surface: 256 and sectors per track: 32. Successive tracks in cylinder can be read without head movement. What is the average access time? [Assume this file is stored in successive sectors and tracks of successive cylinders, starting at sector 0, track 0 of cylinder i]. Find the time taken to transfer an 8 MB file?
5. Given a non-pipelined architecture running at 2.5 GHz, that takes 5 cycles to finish an instruction. You want to make it pipelined with 5 stages. Due to hardware overhead the pipelined design will operate only at 'x' GHz. 30% of instructions cause a stall of 10 cycles each during the IF stage due to I-cache miss. But there is no such stall for non-pipelined architecture. Assume that there are no other hazards. It was observed that the speedup obtained by the pipelined design over the non-pipelined design is 1. What is the value of 'x'?
6. A microprocessor with clock frequency of 500 kHz is connected to an I/O device that employs DMA to transfer data of size 8 bytes with transmitting rate of 2KBps. During the data transfer, DMA uses 15 cycles for initialisation and 3 cycles to transfer one byte of information between memory and I/O device. Find the % time CPU is blocked in burst mode DMA.
